

8. (a) Some elements in the p -block form compounds where the p -block atom does not have eight electrons in its outer shell.

- (i) Explain why nitrogen can only form a chloride with eight outer shell electrons, but phosphorus can form a chloride with a different number of outer shell electrons. [2]

You should give the chemical formulae of relevant compounds in your answer.

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- (ii) Explain why aluminium forms compounds that are electron deficient. Show how one of these compounds can act to gain a full outer shell. [2]

You should give the dot and cross diagrams of the electron deficient species.

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Carbon	There are two common oxides of carbon. Carbon dioxide is an acidic oxide and carbon monoxide can be used as a reducing agent. Both of these are gases with very low boiling temperatures.
Lead	There are two common oxides of lead. Lead(II) oxide is an amphoteric oxide and lead(IV) oxide can be used as an oxidising agent. Both of these are solids that exhibit a large degree of ionic character.

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SECTION B

Answer all questions in the spaces provided.

8. (a) Sodium chloride and sodium bromide both react with concentrated sulfuric acid. Describe the observations in both reactions and explain why they are different. [3]

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- (b) Fluorine can form the weak acid hydrofluoric acid, HF. This acid has a K_a value of $7.20 \times 10^{-4} \text{ mol dm}^{-3}$.

- (i) Write an expression for the K_a of hydrofluoric acid. [1]

- (ii) Calculate the pH of a solution of hydrofluoric acid of concentration $0.100 \text{ mol dm}^{-3}$. [3]

pH =



- (iii) The concentration of a solution of hydrofluoric acid can be found by titrating against sodium hydroxide. Not all acid-base indicators would be suitable for this titration. Explain what features would make an indicator suitable for use in a weak acid-strong base titration. [2]

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- (iv) Addition of 12.5 cm^3 of 0.100 mol dm^{-3} sodium hydroxide to 25.0 cm^3 of hydrofluoric acid of concentration 0.100 mol dm^{-3} forms a buffer solution.

- I. Explain how this buffer solution works. [3]

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- II. Calculate the pH of this buffer solution. [2]

pH =

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SECTION B

Answer **all** questions.

8. The Mohs scale measures the hardness of different materials and runs from 1 to 10, with 10 being the hardest. A fingernail is rated as 2.5 on the Mohs scale so any material that can be scratched with a fingernail has a hardness of less than 2.5. Graphite and the minerals halite and tachyhydrite, can all be scratched with a fingernail.

- (a) The hardness value of graphite is approximately 1.5.

Describe the structure of graphite and explain why it is soft.

[2]

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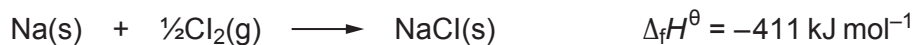
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- (b) Halite is the mineral name for rock salt. It contains sodium chloride.

The equation corresponding to the standard enthalpy change of formation of sodium chloride is as follows.



Some standard enthalpy changes are given in the table.

Enthalpy term	Standard molar enthalpy change / kJ mol ⁻¹
first ionisation energy of sodium	494
electron affinity of chlorine	-364
bond energy of Cl ₂	242
enthalpy of atomisation of sodium	109

- (i) State the enthalpy of atomisation of chlorine.

[1]

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- (ii) Use the standard enthalpy changes to find the standard enthalpy change of lattice formation of sodium chloride. [3]

Standard enthalpy change of lattice formation = kJ mol^{-1}

- (iii) A student suggests that the entropy change for the formation of sodium chloride must be positive as the reaction occurs easily.

Is the student correct? Justify your answer. [2]

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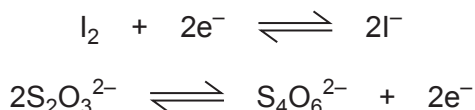
SECTION B

8. When performing redox titrations the concentration of one solution is found by reaction with another solution of known concentration. Some oxidising agents are classed as primary standards as their concentrations can be calculated precisely from the mass used. Other oxidising agents need to be standardised by reaction with solutions of known concentration before use.

- (a) Potassium dichromate, $K_2Cr_2O_7$, is a primary standard. Calculate the concentration of a solution of potassium dichromate made from 5.00 g of potassium dichromate in 250 cm^3 of deionised water. [2]

concentration = mol dm^{-3}

- (b) Sodium thiosulfate, $Na_2S_2O_3$, can be standardised using potassium dichromate. To do this, acidified potassium dichromate is used to oxidise iodide ions to iodine molecules and the sodium thiosulfate is then used to reduce the iodine formed.



- (i) Write the half-equation for acidified dichromate ions acting as an oxidising agent. [1]

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- (ii) Write an equation for the oxidation of iodide ions to iodine molecules using acidified dichromate. [1]

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- (iii) Excess KI(aq) was added to 25.0 cm^3 of the aqueous potassium dichromate from part (a) with a small amount of dilute acid. This produced iodine. Aqueous sodium thiosulfate was added from a burette and 24.30 cm^3 of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ was needed for complete reaction with the iodine.

I. The titration used an indicator. Name the indicator used. [1]

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II. Calculate the concentration of the aqueous sodium thiosulfate. [3]

concentration = mol dm^{-3}



- (c) Addition of aqueous sodium hydroxide to the solution of potassium dichromate causes a colour change.
State the colour change and explain why this is not a redox reaction. [2]

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- (d) Heating solid potassium dichromate can cause it to decompose releasing oxygen gas.



$$\Delta H^\theta = 348 \text{ kJ mol}^{-1} \quad \Delta S^\theta = 385 \text{ J K}^{-1} \text{ mol}^{-1}$$

Substance	Standard enthalpy change of formation, $\Delta_f H^\theta / \text{kJ mol}^{-1}$
K_2CrO_4	-1382
Cr_2O_3	-1128

- (i) Calculate the standard enthalpy change of formation of potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$. [2]

$$\Delta_f H^\theta = \dots\dots\dots \text{kJ mol}^{-1}$$

- (ii) Calculate the minimum temperature required for this reaction. [2]

$$\text{minimum temperature} = \dots\dots\dots \text{K}$$

